LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART III

SESSION 11

Exchange Rates, Capital Flows & the Impossible Trinity

From markets, to macro, to decisions that matter — in Nigeria and the world.



Reading: Thailand: An Imbalance of Payments (209-021-1) · Krugman, Obstfeld & Melitz, International Economics · Facilitator: Elvis Ozegbe

RECAP

Part II in Three Ideas



Shocks & inflation

Supply shocks (FX, fuel, food) raised Nigerian prices; expectations decide whether inflation sticks.



Policy levers

The CBN works through rates, liquidity and the balance sheet; fiscal deficits financed by the central bank undermine it.



Fragility

Higher rates hit bond values and bank capital: duration gaps turn rate shocks into capital losses.

TODAY'S PLAN

Session 11 at a Glance — 75 Minutes

Why the naira moved from ₦460 to ₦1,500 per dollar — and what every importer, exporter and borrower must learn from it.



Recap & Hook

One dollar, two prices

Regimes & the Trinity

Pegs, floats, controls and the policy you must give up

Flows & Sudden Stops

The BoP, hot money, carry trades and currency mismatch

Lab: Reading the Naira

Premium, current account and a dollar loan

BY THE END OF TODAY

Session 11 Learning Objectives

You will be able to:

01

Explain FX Regimes

Compare pegs, managed floats and free floats, and explain the impossible trinity in plain language.

02

Read the Balance of Payments

Trace where Nigeria's dollars come from and go to, and spot the warning signs of a sudden stop.

03

Manage Currency Risk

Identify currency mismatch in a business and explain how a weaker naira hits costs, debt and competitiveness.

THE HOOK

One Dollar, Two Prices

“We buy dollars at the official rate. Our competitor pays the parallel rate. We have a cost advantage — until we don't get any dollars at all.”

For most of 2016–2023 Nigerian firms lived with this. Three questions:

Why two prices?

If the official price is held below what the market will pay, demand exceeds supply and dollars must be rationed.

Who gets the dollars?

Rationing created queues, backlogs and a valuable privilege for whoever had access.

What happens at unification?

When rationing ends, the official rate jumps toward the market rate — a big, one-off devaluation.

NIGERIAN BUSINESS CASE

From ₦460 to ₦1,500: Lessons from the Naira's Unification

The situation

Until mid-2023 Nigeria kept an official rate near **₦460/$** while the parallel market traded far weaker. In June 2023 the CBN unified its FX windows; by 2024 the naira traded around **₦1,500/$**.

The decision question

Why did the old system break, who gained and lost from unification, and what should a Nigerian firm do differently now?

The data we have

ng_macro_monthly.csv — official and parallel rates, reserves, policy rate, inflation. ng_bop_quarterly.csv — trade, remittances, services, FDI and portfolio flows, 2015–2025. *Stylised teaching data.*

What a good answer looks like

A clear account of the **regime choice** using the impossible trinity, plus a **currency-risk checklist** for a firm.

PART

01

Regimes and the Impossible Trinity

You can have two of three — never all three.

What Sets the Exchange Rate?

IN PLAIN ENGLISH

An exchange rate is a price: naira per dollar. It is set by **who wants dollars** (importers, students abroad, investors leaving, people saving in dollars) and **who supplies them** (oil exporters, remittances, foreign investors coming in, the CBN).

EXAMPLE

When oil earnings fell in 2015–16 and again in 2020, dollar supply dropped. The CBN either let the naira fall or spent reserves and rationed dollars.

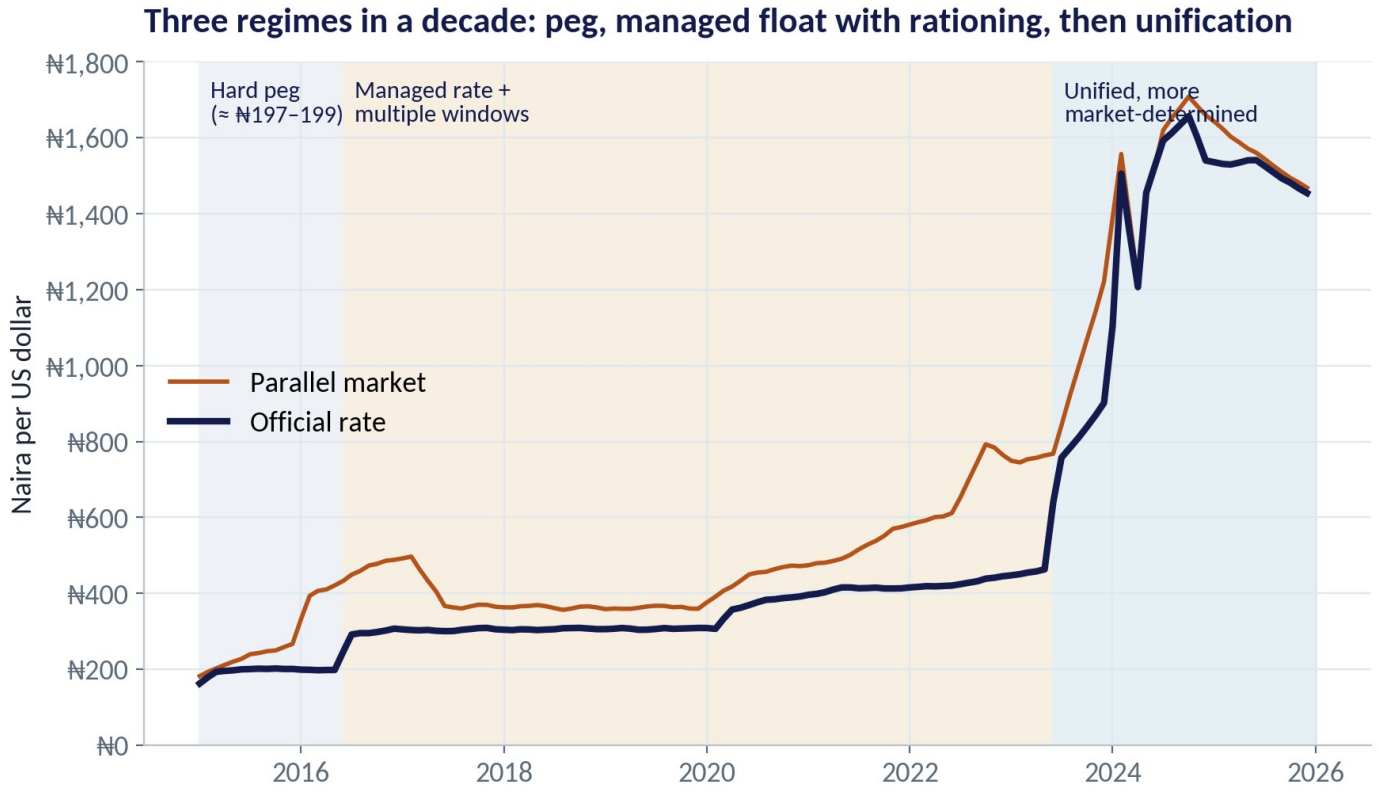
THE FORMULA

dollar demand > dollar supply → naira weakens (or reserves fall)

WHY A MANAGER CARES

Every forecast of the naira is really a forecast of dollar inflows and outflows. Learn to read those flows and you will rarely be surprised.

A Decade of Naira Policy



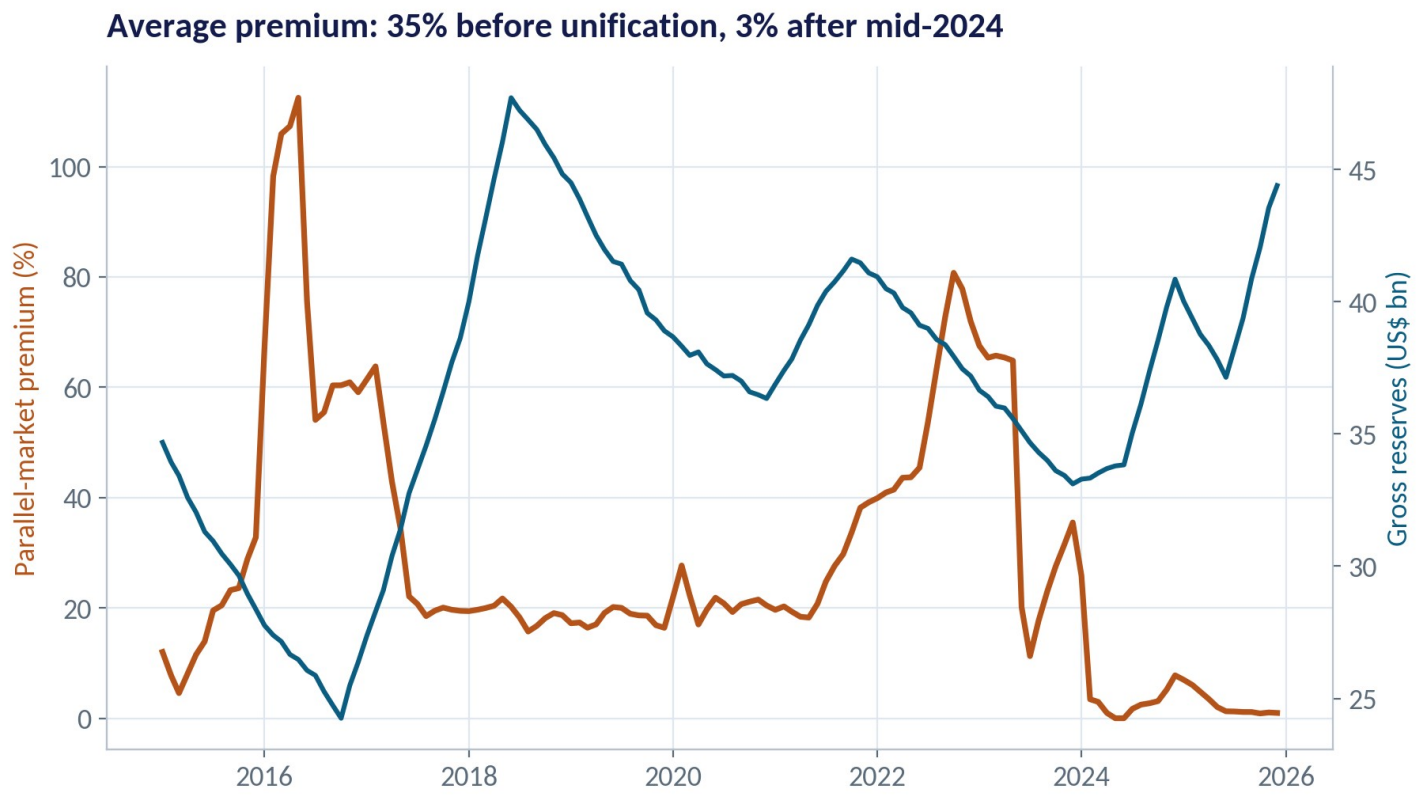
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S11-F01 · replicate: s11_figures.R / .py

WHAT TO SEE

- **2015–mid-2016**: a near-fixed rate around ₦197–199 per dollar.
- **2016–2023**: a managed rate with several official windows, and a wide parallel market.
- **From June 2023**: windows unified; the rate became much more market-determined.
- Each switch came after the old regime ran short of dollars.

The Premium Measures Rationing



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S11-F02 · replicate: s11_figures.R / .py

WHAT TO SEE

- The parallel premium averaged about **35%** before unification and peaked above **100%** in 2016.
- Since mid-2024 it has averaged under **3%.**
- A high premium tells you official dollars are being **rationed**.
- Falling reserves alongside a rising premium is the classic warning of a coming devaluation.

THE CORE IDEA

The Impossible Trinity

IN PLAIN ENGLISH

A country would like three things: a **stable exchange rate**, **free movement of capital**, and **its own interest-rate policy**. It can only have **two** at a time.

EXAMPLE

If the CBN fixes the naira and lets capital move freely, it must set interest rates to defend the peg — it loses control of domestic policy. Nigeria before 2023 kept the peg and its own rates by **rationing** dollars — a form of capital control.

THE FORMULA

fixed rate + free capital + own monetary policy = impossible

WHY A MANAGER CARES

When a government promises all three, one promise will break. For a business, the question is **which one, and when**.

Pick Any Two Corners

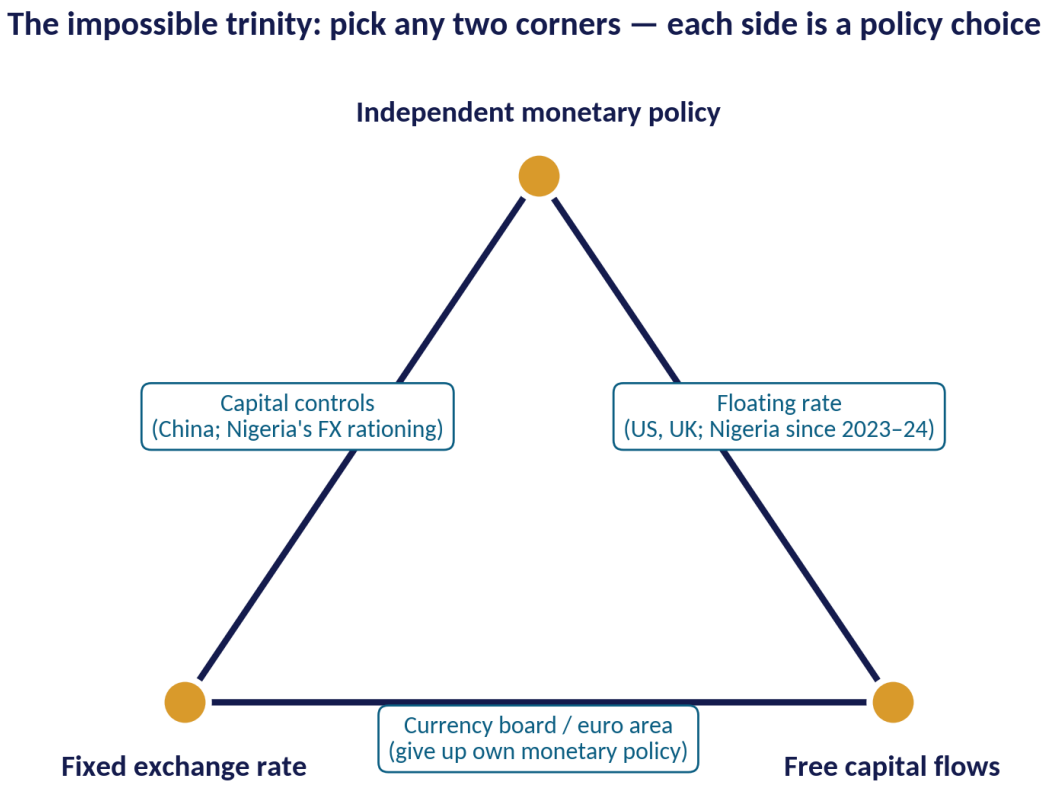


Fig S11-F05 · replicate: s11_figures.R / .py

WHAT TO SEE

- **Currency board / monetary union**: fixed rate and open capital, but no independent rates (e.g. the euro area, the CFA franc zone).
- **Capital controls**: fixed or managed rate with own policy — at the cost of rationing and parallel markets.
- **Floating rate**: open capital and own policy — the exchange rate does the adjusting.
- Nigeria moved from the controls side towards the floating side in 2023–24.

CONCEPT CHECK

Quick Check: What Would You Say?

?

A country fixes its currency, allows investors to move money freely, and wants to cut interest rates to boost growth. What is most likely to happen?

A. Growth rises and the peg holds

B. Investors move money out to earn higher returns abroad, reserves fall, and the peg comes under pressure

C. Inflation falls automatically

D. Nothing changes because the rate is fixed

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

02

Flows, Hot Money and Sudden Stops

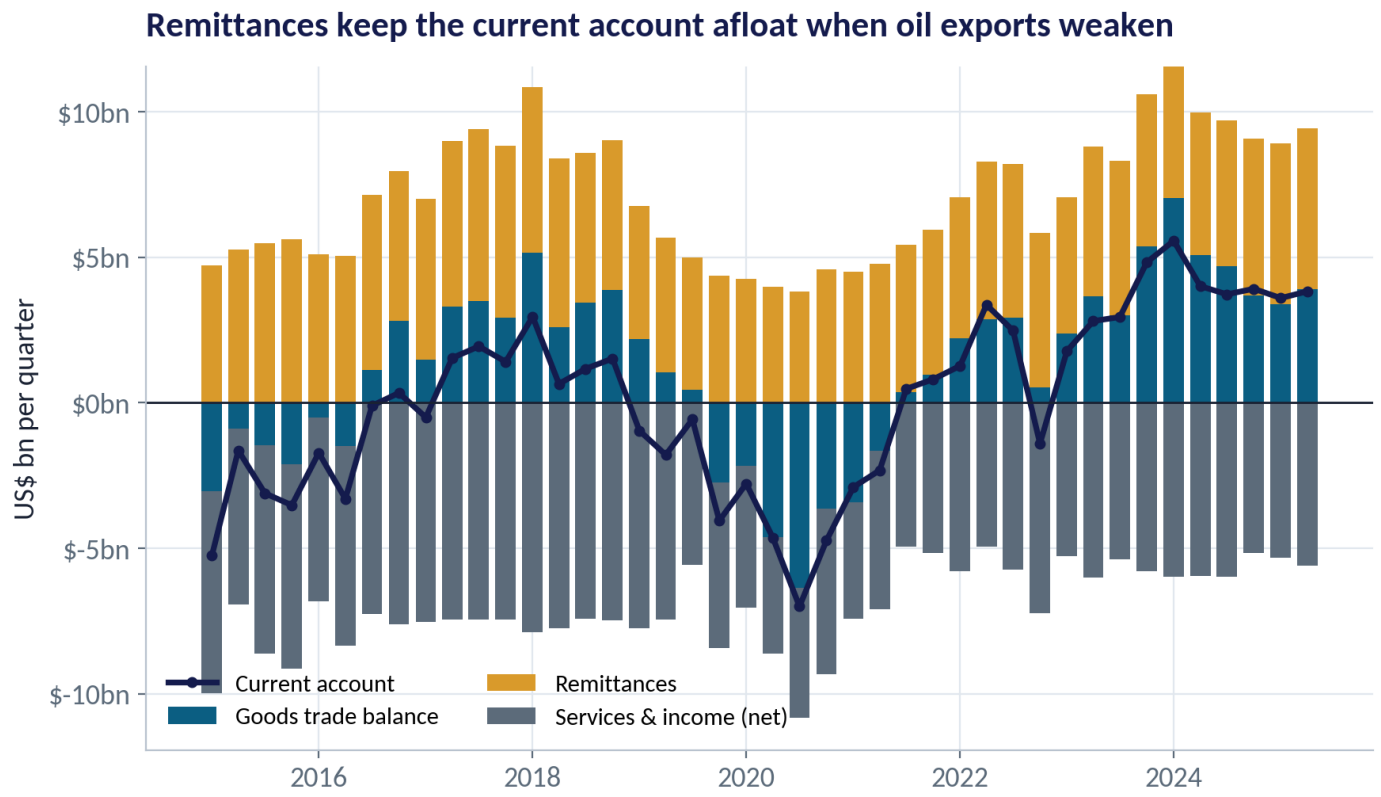
Where the dollars come from — and how fast they can leave.

Nigeria's Dollar Account in Plain English

Line	What it records	Nigeria's pattern
Goods trade balance	Exports minus imports of goods	Swings with oil: surplus when oil is high, deficit when it falls
Services & income	Freight, travel, school fees, profits and interest paid abroad	Persistently in deficit
Remittances	Money sent home by Nigerians abroad	Large and steady — about US$5bn a quarter in our data
FDI	Long-term investment in factories and firms	Small and falling in our data
Portfolio flows	Foreign money in bonds, bills and shares	Large and volatile — 'hot money'

Current account = trade + services & income + remittances. Financial flows must cover any current-account deficit — or reserves fall.

Remittances Keep the Account Afloat



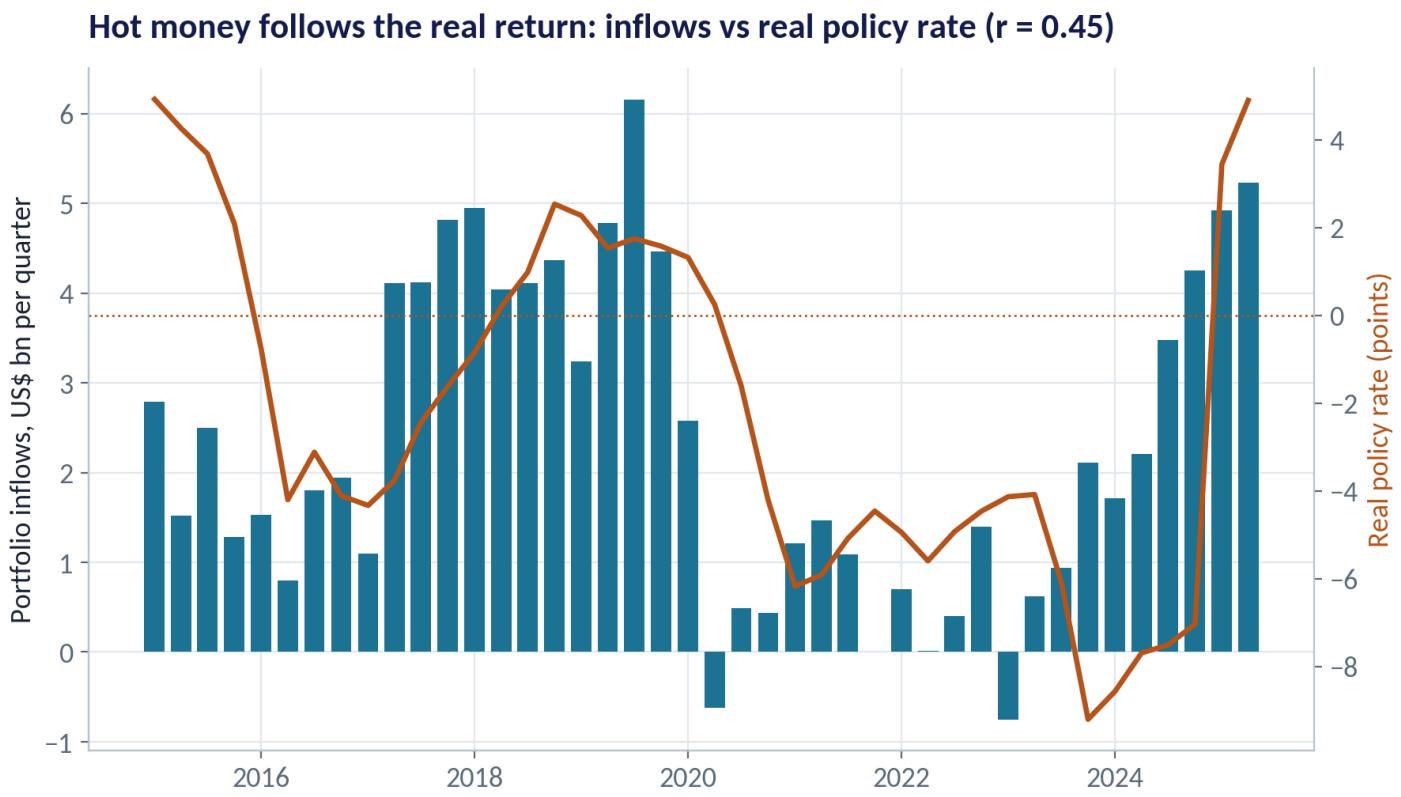
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S11-F03 · replicate: s11_figures.R / .py

WHAT TO SEE

- Remittances (gold) average about **US$5.1bn** a quarter — a stable dollar source.
- Services and income (grey) are a steady drain.
- The current account turns negative when oil weakens: about **−US$4.6bn** in 2020Q2.
- Higher oil and a weaker naira restored surpluses in 2023–24.

Portfolio Flows Chase the Real Return



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S11-F04 · replicate: s11_figures.R / .py

WHAT TO SEE

- Foreign portfolio investors come when the **real** return — interest minus inflation, adjusted for expected depreciation — is attractive.
- In our data the correlation between inflows and the real policy rate is **0.45**.
- These flows can reverse within weeks: a **sudden stop**.
- A central bank that relies on hot money to fund the peg is building a trap.

Borrow Cheap, Lend Dear — Until the Currency Moves

IN PLAIN ENGLISH

In a **carry trade**, investors borrow in a low-interest currency (dollars) and buy high-yield naira bills. They earn the interest gap — **as long as the naira does not fall** by more than that gap.

EXAMPLE

A 20% naira bill against a 5% dollar cost earns about 15% — but a 40% devaluation turns that into a large loss.

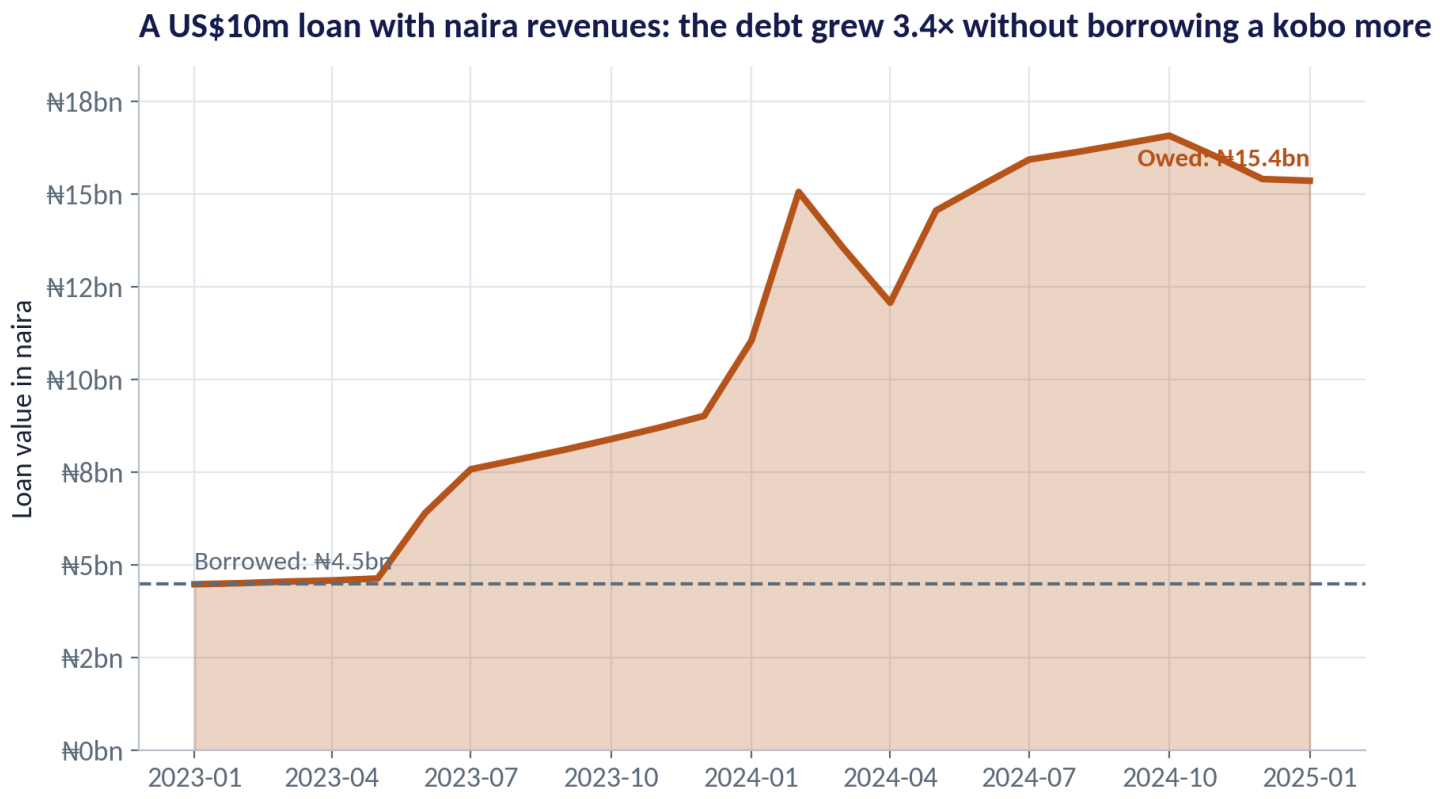
THE FORMULA

$$\text{carry return} \approx \text{naira yield} - \text{dollar yield} - \text{naira depreciation}$$

WHY A MANAGER CARES

When carry investors all head for the exit at once, the naira falls fast and reserves drain. Watch their positioning, not just the interest rate.

The Dollar Loan That Tripled



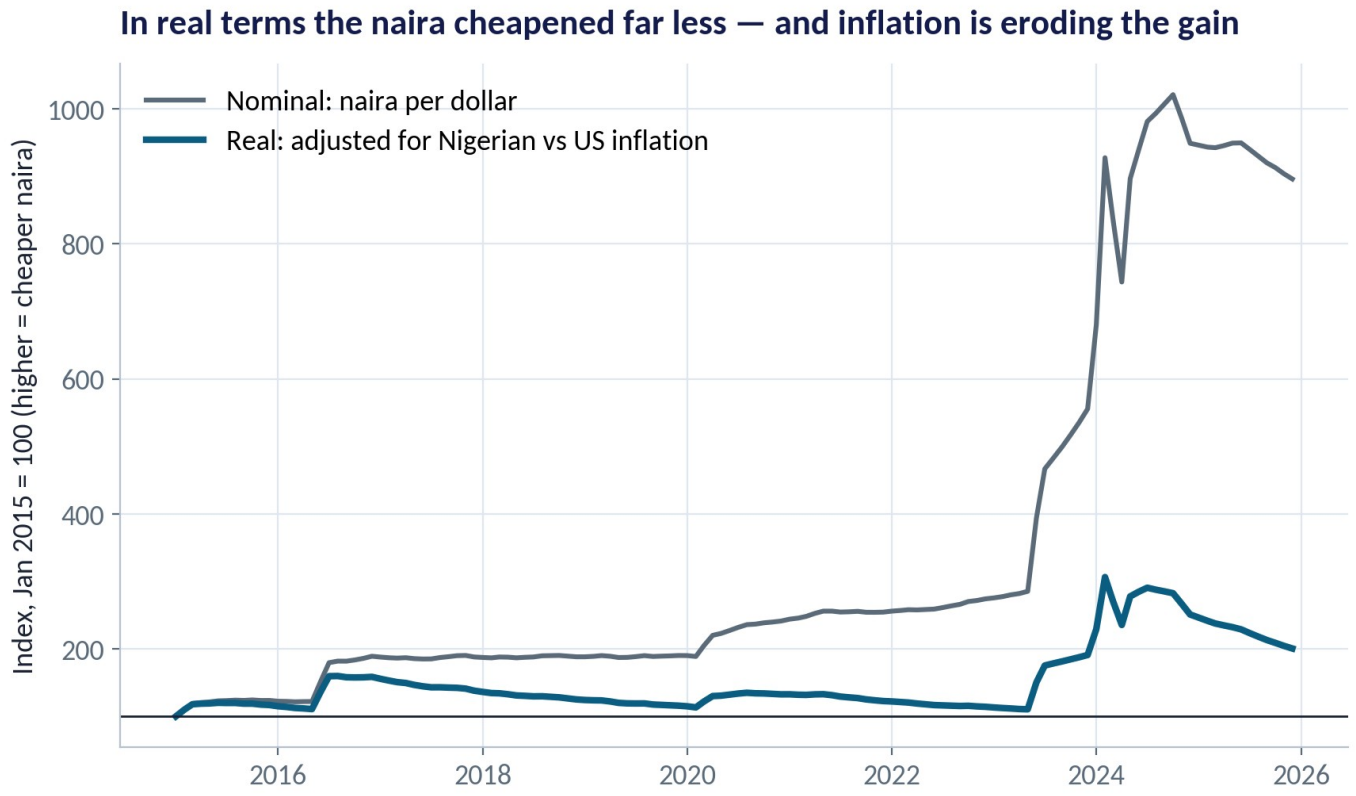
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S11-F06 · replicate: s11_figures.R / .py

WHAT TO SEE

- A firm borrows **US$10m** in January 2023 to fund naira-earning operations: **₦4.5bn**.
- By January 2025 the same debt is worth **₦15.4bn**.
- Nothing changed in the business — only the exchange rate.
- Mismatch, not the dollar itself, is the danger: borrow in the currency you earn.

Did the Naira Really Get Cheaper?



Real rate uses stylised CPI with US inflation assumed at 2.5% a year. EEGB stylised teaching data.

Fig S11-F07 · replicate: s11_figures.R / .py

WHAT TO SEE

- The **real** rate adjusts the nominal rate for inflation at home and abroad.
- From 2016 to 2023 the real index drifted from about **159 to 110**: high inflation made the naira **overvalued** at a fixed rate.
- Unification produced a sharp real depreciation (peak about **306** in Feb 2024).
- Since then inflation has eroded part of the gain (about **201** by end-2025).

How Firms Get Caught by the Currency

THE TRAP	THE BETTER HABIT
✗ Planning on the official rate when you actually buy at another rate.	✓ Budget at the rate you can **really** access, plus a stress case.
✗ Borrowing in dollars because the interest rate is lower.	✓ Compare **all-in cost including expected depreciation**; avoid mismatch.
✗ Treating a long-stable peg as permanent.	✓ Watch **reserves, the premium and the current account**: they warn early.
✗ Assuming a weaker naira automatically makes exports competitive.	✓ Check the **real** exchange rate: inflation can erase the gain.
✗ Ignoring hot-money positioning.	✓ Large foreign holdings of short-term naira bills are a **sudden-stop** risk.

CONCEPT CHECK

Quick Check: What Would You Say?

?

Your company borrowed US$10m in January 2023 at 8% a year instead of ₦4.5bn at 18%. Using the stylised rates, what is the main lesson two years later?

A. The dollar loan was cheaper because 8% < 18%

B. The naira value of the dollar debt rose from about ₦4.5bn to ₦15.4bn, wiping out the interest saving many times over

C. The two loans cost the same

D. Exchange rates don't matter for debt

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Lab: Reading the Naira

YOU WILL NEED

Data

`data/ng_macro_monthly.csv`
`data/ng_bop_quarterly.csv`

R script

`sessions/s11/R/s11_lab.R`

Python script

`sessions/s11/python/s11_lab.py`

20 minutes

1. Compute the **parallel premium** by year and compare it with reserves.
2. Rebuild the **current account** from its parts, by year.
3. Correlate **portfolio inflows** with the real policy rate.
4. Track a **US$10m loan** in naira from 2023 to 2025.
5. Place Nigeria's pre- and post-2023 regimes on the **impossible trinity**.

Deliverable: a one-paragraph note to your CFO: three signals that the naira is under pressure, and where to find them.

WORDS TO OWN

Today's Vocabulary in Plain English

Exchange-rate regime	The rules for how a currency's value is set: fixed, managed or floating.
Peg	A currency held at a fixed rate against another.
Parallel premium	How much weaker the parallel rate is than the official rate.
Impossible trinity	A fixed rate, free capital and own monetary policy cannot coexist.
Current account	Trade, services, income and transfers with the rest of the world.
Remittances	Money sent home by citizens working abroad.
Hot money	Short-term portfolio flows that move quickly in and out.
Sudden stop	An abrupt halt or reversal of capital inflows.
Carry trade	Borrowing in a low-rate currency to invest in a high-rate one.
Currency mismatch	Debts in one currency, revenues in another.

KEY TAKEAWAYS

Session 11 in 5 Ideas

01 The exchange rate is a **price set by dollar inflows and outflows**.

02 The **impossible trinity**: a country must give up one of a fixed rate, free capital or its own policy.

03 A wide **parallel premium** means rationing; it usually ends in devaluation.

04 **Hot money** can leave fast; remittances are Nigeria's steadiest dollar source.

05 For firms, the killer is **currency mismatch** — borrow in the currency you earn.



NEXT › Session 12: Trade & Industrial Policy

Before next session: Read Sáenz & Revilla, 'Creating More Resilient Supply Chains' (MIT Sloan Management Review). List your company's three most important imported inputs and where they come from.